

**DC and AC Characterization of BJT Amplifiers (Common Collector and
Common Emitter)**

First Formal Report for Understanding BJT Amplifiers
ECE 333-001 - Electronics 2

Submitted By

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Abstract

This laboratory experiment investigated the AC and DC characteristics of bipolar junction transistor (BJT) amplifiers using common-emitter and common-collector configurations. In part one a common-collector circuit was designed to observe operating regions and current gain. Both a 2N2222 and 2N3904 transistor were used to observe these characteristics. Measurements demonstrated the transition from forward active to saturation mode as well as the relationships between base current and base-emitter voltages. Current gain was compared between the two different transistors to reveal the difference between the transistors. In the second part of the lab, a common-emitter circuit was designed to ensure the q-point was near the mid-supply of the DC voltage to show effective signal amplification. More experiments were conducted to observe phase shift and signal clipping. The measured data from this experiment closely follow the expectations expected from this experiment.

Table of Contents

	Page
INTRODUCTION	1
OPERATING REGIONS AND BIASING OF A BJT	2
COMMON-COLLECTOR CIRCUIT DESIGN	3
COMMON-COLLECTOR RESULTS AND ANALYSIS	5
COMMON-EMITTER CIRCUIT DESIGN	8
COMMON-EMITTER RESULTS AND ANALYSIS	10
OBSERVATIONS AND DISCUSSION	13
CONCLUSION	14
REFERENCES	15
APPENDICES	16

INTRODUCTION

The objective of the laboratory was to design and analyze a common-collector and common-emitter circuit in order to better understand the difference between direct current (DC) and alternating current (AC) operation of a BJT amplifier. In the first part of this lab a common-collector circuit was designed and analyzed to investigate the characteristics of this specific type of circuit. Two different NPN transistors were used to analyze these behaviors; a 2N2222 and a 2N3904 transistor. To properly analyze the characteristics of these different circuits, values were recorded for base-emitter and collector-emitter voltages as well as the base and collector currents. These values were used to determine the current gain and observe the transition from forward active mode to saturation mode. In the second part of this lab, a common-emitter amplifier was designed so that at the Q-point the transistor is in forward active mode. Measurements were taken to show the voltage gain, examples of clipping, frequency response, and phase shift. The main goal of this report is to show the method of designing a BJT circuit, measurements of desired values, graphically representing these values, and to compare the results of both configurations. This shows a bigger picture of how BJT's operate as well as the characteristics of different setups.

OPERATING REGIONS AND BIASING OF A BJT

To better understand what this lab experiment is covering it can be helpful to understand some background information about the components and their behavior. A bipolar junction transistor, commonly referred to as a BJT, is a three terminal semiconductor device that operates as a current-controlled current source. The behavior of a BJT is dependent upon the base-emitter and collector-emitter voltages (V_{be} & V_{ce}). These voltages will determine the operating region of the BJT.

There are three different operating regions: cutoff, forward active, and saturation. In the forward active mode, the collector-base junction is forward biased with a voltage of around 0.7 volts and the collector-base junction is reverse biased with a voltage that can be found through measurements (no specific target voltage). In FAM (Forward Active Mode), the current from the base is proportional to the collector current from the identity ' $I_c = \beta I_b$ ', where the β (beta) is the current gain. In saturation mode, both the base-emitter and the collector-base junctions are both forward biased. In this mode the collector-emitter voltage drop is significantly smaller (around 0.2 - 0.3 volts) and the base-emitter voltage drop is again around 0.7 volts. A big difference between forward active and saturation is that in saturation the identity ' $I_c = \beta I_b$ ' no longer holds true. In cutoff mode, the name sounds exactly what you would expect and the BJT produces a negligible current due to neither of the junctions being forward biased.

In order to get an effective amplification (important for part two of this experiment), the transistor must be properly biased at a Q-point. This Q-point is the operating condition whenever there is no AC signal present. This proper biasing will ensure that the input signal will produce a proportional output signal without changing the transistors region to cutoff or saturation. Without proper biasing the output signal could experience clipping or distortion. Understanding these

concepts is crucial for interpreting the AC and DC characteristics observed throughout this experiment.

COMMON-COLLECTOR CIRCUIT DESIGN

In order to start understanding the characteristics of a common-collector circuit, there first must be a designed circuit to experiment with. The circuit schematic is given from the lab manual and is shown below in **Figure 1**.

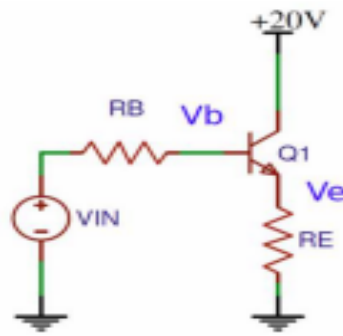


Figure 1. BJT Common-Collector Schematic

The overall layout of the circuit is clearly shown, so all that is left to “design” are the two resistors R_B and R_E . The range of voltages are given through the lab manual and will make designing the resistors relatively simple. The voltage ranges from 4 volts to 24 volts. To calculate the resistors required to show both forward active and saturation modes some key equations are necessary to identify viable resistances. These equations are used to calculate appropriate resistor values and can be found below in **Figure 2**.

$$\begin{aligned}
 \textcircled{1} \quad I_B &= \frac{V_{in} - V_B}{R_B} \\
 \textcircled{2} \quad V_B &= V_E + V_{BE} \quad V_{BE} \approx 0.7 \text{ V} \\
 \textcircled{3} \quad I_E &= \frac{V_E}{R_E} \\
 \textcircled{4} \quad I_C &= \beta I_B \\
 I_E &\approx (\beta + 1) I_B \\
 \textcircled{5} \quad I_E &= \frac{(\beta + 1)(V_{in} - V_{BE})}{R_B + (\beta + 1)R_E}
 \end{aligned}$$

Figure 2. Important Common-Collector Equations

Using these equations resistance values can be determined that will keep the transistor in forward active mode as well as push past into saturation. In order to do this there must be a few assumptions made. Firstly, the voltage across the base-emitter junction is roughly 0.7 volts. Secondly, the current gain (or beta) is around 100. Now to solve for the emitter current as a function of the input voltage. The equation that was derived is as follows: ‘ $V_E(V_{in}) = I_E R_E$ ’. This can be used to solve for an R_B value that will ensure the transistor is on at 4V. Notice the R_E value is still unknown in this equation it is required to find that first. To find R_E it is required to use the emitter current formula (from **Figure 2**). By replacing the collector-emitter voltage with a typical saturation value the resistance value that would satisfy saturation can be found. After calculations, the value for R_E came out to be 4.5kOhm. Now that R_E is found the next step is to solve for R_B . Going back to the derived formula and plugging in the minimum input voltage value, the resistance value came out to be 44,500 ohms. For simplicity sake the value was rounded to 44 kOhms. To verify that the transistor will enter saturation, the formula (5) from **Figure 2** can be used. Plugging in values to solve for V_E the result was 21.2 volts. Since this voltage is higher than the input voltage it implies that the transistor is indeed in saturation mode. The final resistance values can be found below in **Table 1**.

Re	Rb
4.5kOhm	44kOhm

Table 1. Calculated Resistance Values (Common-Collector)

It is important to note that these resistance values are used in both circuit configurations. The same setup that was used for the 2N2222 transistor is also used for the 2N3904 experiments.

COMMON-COLLECTOR RESULTS AND ANALYSIS

After the process of designing and assembling the common-collector circuit onto a breadboard, measurements are required to fully analyze the transition from forward active to saturation mode. Following the schematic given from the lab manual, it is easy to notice that there is only a DC input voltage. This means that the only tools needed to complete this experiment are a DC voltage source and a multimeter to measure the voltage values. The currents are found through the application of Ohm's law by dividing the voltage drops by their corresponding resistor values. Following these methods, the following **Table 2** shows all recorded data from the 2N2222 transistor configuration.

V _{in}	V _{rb} (V)	V _{re} (V)	V _{ce} (V)	V _{be} (V)	I _c (mA)	I _b (uA)	I _c /I _b
4 V	0.28	3.09	16.9	0.647	0.681	6.36	106.9
8 V	0.531	6.82	13.17	0.641	1.504	12.07	124.6
12 V	0.76	10.58	9.4	0.65	2.334	17.27	135.1
16 V	0.99	14.3	5.64	0.656	3.156	22.5	140.2
20 V	1.22	18.09	1.88	0.646	3.992	27.73	144
24 V	3.55	19.77	0.115	0.648	4.312	80.68	53.45

Table 2. 2N2222 Common-Collector Measurements

This data can be used to represent the current gain (β) as well as show exponential increase of base current as the base-emitter voltage approaches the turn on voltage. Using graphing software **Figure 3** and **Figure 4** represent this.

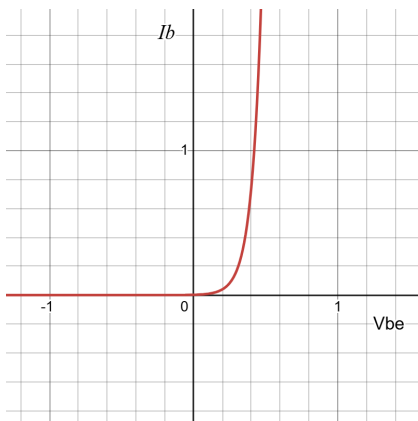


Figure 3. Ib vs. Vbe (2N222)

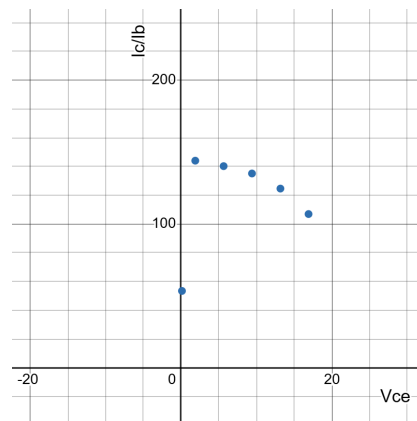


Figure 4. Ic/Ib vs Vce (2N2222)

These graphs show (almost) exactly what was expected going into this experiment. **Figure 3** shows the exponential increase in base current as the voltage reaches the turn on voltage of around 0.65 volts. In **Figure 4**, current gain is not very constant, but the dramatic drop at around 0.1 volts still represents saturation that was expected.

Seeing as the results from the 2N2222 transistor match with the expectations of this experiment, now we can take a look at the 2N3904 to try and observe any differences between the two. Shown below in **Table 3** is the data collected from the 2N3904 configuration.

V _{in}	V _{rb} (V)	V _{re} (V)	V _{ce} (V)	V _{be} (V)	I _c (mA)	I _b (uA)	I _c /I _b
4 V	0.094	3.28	16.7	0.624	0.727	2.14	340
8 V	0.2	3.28	12.83	0.638	0.727	4.55	159
12 V	0.308	11.03	8.94	0.649	2.444	7	349
16 V	0.425	14.76	5.06	0.661	3.27	9.66	338
20 V	0.55	18.76	1.2	0.673	4.16	12.5	333
24 V	3.43	19.8	0.088	0.685	4.32	78	55

Table 3. 2N3904 Common-Collector Measurements

Using this data we can again represent the current gain (β) as well as the increase in base current graphically. The following **Figure 5** and **Figure 6** show the resulting graphs.

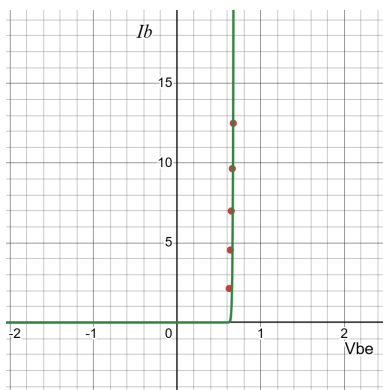


Figure 5. I_b vs. V_{be} (2N3904)

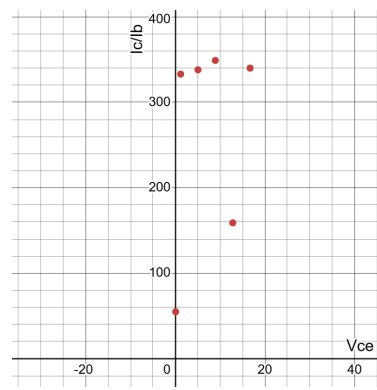


Figure 6. I_c / I_b vs. V_{ce} (2N3904)

These graphs again show what we are looking for, but with a noticeable difference.

Figure 5 shows the exponential current as it reaches the turn on voltage as expected. However, in **Figure 6** there is a noticeable outlier in the graph. It appears that the current gain takes a dramatic drop before reaching saturation. Looking back at the data table, this can be accounted for by noticing the misinput while taking data. The collector current and voltage drop values seem to be accidentally copied from the previous measurement, affecting the resulting graph.

Looking at the differences between the 2N2222 and the 2N3904 configurations it is immediately apparent that the current gain is a lot higher in the 2N3904 compared to the 2N2222. This makes sense when looking at the datasheets for both the 2N2222 and the 2N3904, as the 2N3904 is rated for a current gain between 100 and 300+. This is because the 2N3904 is designed for small signal amplification while the 2N2222 is a more general-purpose transistor. The larger current gain is more beneficial for small-signal gain.

COMMON-EMITTER CIRCUIT DESIGN

For this second part in the lab, a common-emitter circuit needs to be designed so that at the Q-point the transistor has a collector voltage of around 10 volts and the base current is approximately 100 times lower than the current flowing in R1. The circuit schematic given in the lab manual will certainly help in the design and can be found below in **Figure 7**.

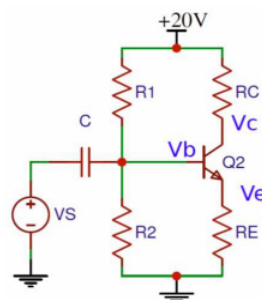


Figure 7. BJT Common-Emitter Circuit Schematic

Some key equations to solving for these resistor values can be found below in **Figure 8**. To accomplish this, the first step is to choose a reasonable collector current to aim for. Then using the first equation from **Figure 8** it is possible to solve for R_C . Choosing R_E can be done by assuming I_E and I_C are close in value, then using the second equation from **Figure 8** you can choose a reasonable R_E value. To choose resistor values for R_1 and R_2 it is required to ensure the base voltage is around the target value of around 3 volts.

$$\begin{aligned}
 &\textcircled{1} \quad R_C = \frac{V_{CC} - V_{CQ}}{I_{CQ}} \\
 &\textcircled{2} \quad R_E = \frac{V_E}{I_{EQ}} \\
 &\textcircled{3} \quad \frac{R_2}{R_1} \approx \frac{V_B}{V_{CC} - V_B} \\
 &\quad \text{Set } V_B \approx V_E + 0.7 \text{ V.} \\
 &\textcircled{4} \quad \frac{V_{CC} - V_B}{R_1} > 10 I_{BQ}
 \end{aligned}$$

Figure 8. Important Common-Emitter Equations

After choosing the resistor values the lab manual requires a verification of the divider current being much larger than the base current. This can be accomplished by using the target base voltage and the supplied 20 volts to show that the To verify the q-point voltage is around 10 volts a thevenin check can be done. Using the calculated resistance values the expected collector voltage is around 9.1 volts. The chosen resistor values can be found below in **Table 4**.

R1	R2	Rc	Re
8kOhm	1.5kOhm	4.56kOhm	1kOhm

Table 4. Calculated Resistor Values (Common-Emitter)

COMMON-EMITTER RESULTS AND ANALYSIS

After designing and assembling the second circuit for this lab, an initial measurement of DC voltage values needs to be recorded in order to ensure that the transistor is in forward active mode. This can be confirmed by measuring the voltage across the base-emitter junction. Since this measurement needs to be only DC, the capacitor is removed to record the values. Below in **Table 5** shows the recorded base-emitter voltage as well as the collector-emitter voltage.

V _{beq} (V)	V _{ceq} (V)
0.658	4.68 V

Table 5. Forward Active Voltage Measurements

Now that the operating region has been verified as forward active, the 0.1 μ F capacitor is added back into the circuit to monitor the AC voltages at the input and the collector terminal. In order to supply AC voltage a waveform generator is applied into the capacitor with a DC voltage still being supplied through R₁ and R_c. The AC signals are monitored through an oscilloscope to easily identify the amplification that is expected. In order to better understand the results many different frequencies and amplitudes were recorded to visualize how the output changes. The frequencies tested range from 100 hertz to 10000 hertz by multiples of 10. Each set of frequencies was tested with a different amplitude, starting at 0.1 volts up to 0.5 volts and finally 1.5 volts. These were chosen to properly show the different cases of saturated and non-saturated. The recorded amplitudes can be found below in **Table 6**.

Freq (Hz)	Amp (V)	Vs (V)	Vc (V)
100	0.1	0.225	0.15
1000	0.1	0.229	0.8
10000	0.1	0.201	1.03
100	0.5	1.01	0.65
1000	0.5	1.05	3.9
100	1.5	2.73	2.01
1000	1.5	2.69	9.6

Table 6. Peak-To-Peak Data Table

The table shows the peak-to-peak voltages of the input signal as well as the signal produced at the collector terminal. The output signals that showed signs of clipping are highlighted. This data table shows exactly what was expected as the collector voltage can be seen to hover around 10 volts (9.6 volts) at the q-point initially designed. The manual requires the small signal gain of the unclipped cases. Below, **Table 7** shows the calculated small signal gain of the 5 unclipped cases.

Freq (Hz)	Amp (V)	Av (Vc/Vs)
100	0.1	0.667
1000	0.1	3.49
10000	0.1	5.12
100	0.5	0.644
1000	0.5	3.71

Table 7. Small Signal Gain (Calculated)

The small signal clearly increases as the frequency increases which is exactly what was expected. This is due to the coupling capacitor it acts as a doorway for the AC signal with the smaller the signal the less “open” it is and vice versa. To better visualize what an unclipped case looks like see **Figure 9** and **Figure 10** below. Both of these readings are unclipped and out of phase by around 140° . This is a bit lower than the expected 180° but still showcases the increased collector current, which in turn increases the voltage drop across the collector resistor and produces the inverted waveform output.

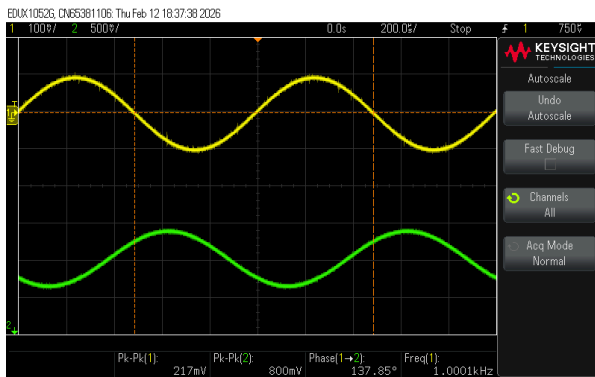


Figure 9. Vc Non-Clipping

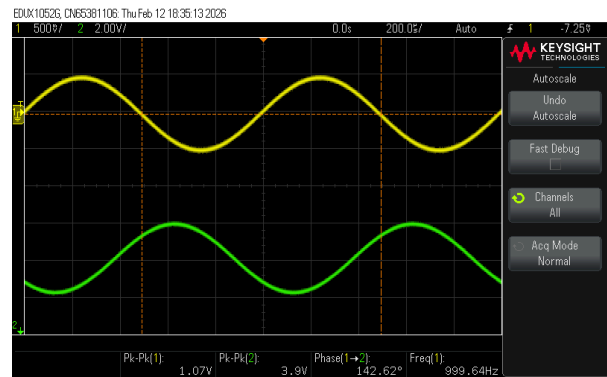


Figure 10. Vc and Vs Out of Phase

When increasing the input signal past a certain point the transistor leaves forward active mode and begins to saturate. The saturation can be visualized by the clipping of the signal. An example of this can be seen below in **Figure 11**.

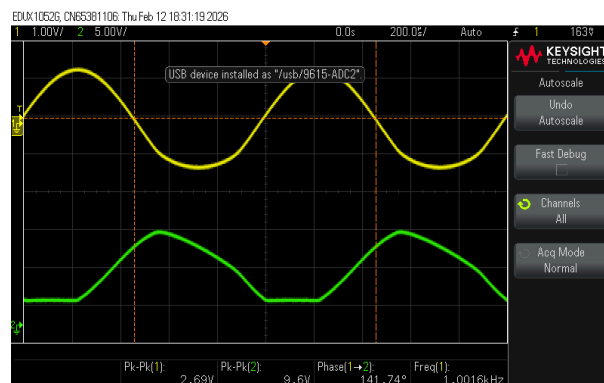


Figure 11. Vs Clipping

While sweeping through frequencies it is helpful to find the corner frequency for experimental purposes. This frequency is found by finding the signal gain that is around 70% of the maximum gain. The corner frequency from this experiment was found to be at the 1000 hertz and 0.1 volt peak-to-peak. This frequency produced a gain of 3.79 which is 74% of the maximum gain found of 5.12.

EXPLANATIONS AND DISCUSSION

The results obtained in this experiment closely align with the theoretical expectations for BJT operations, with a few discrepancies from human/instrumental error. In the first part of the lab the transition from forward active to saturation mode was clearly demonstrated as the collector-emitter voltage approached saturation levels. The relationships between base current and base-emitter voltage as well as current gain were observed through graphical representation. A notable observation from the first part is how the different transistors produce different magnitudes of current gain. With the 2N2222 maintaining around 100+ gain, and the 2N3904 maintaining around 300+ gain. This shows the importance in selection of components during the design process.

In the second part of this lab proper biasing through design was confirmed through experimentation. The measured collector voltage of 9.6 V confirms the q-point was designed properly. Small signal gain was measured from collected oscilloscope data and analyzed to determine the operation with an AC input signal. The smaller frequencies produced a smaller gain and the higher frequencies produced a higher gain. This was observed to show the capacitor impedance. Clipping was observed at higher amplitudes due to the BJT entering saturation mode. The phase shift was also observed, further confirming the expected common-emitter operation.

CONCLUSION

This laboratory experiment successfully demonstrated both DC and AC characteristics of BJT amplifiers using both a common-emitter and common-collector configuration. The designed common-collector circuit represented the relationships between base current and base-emitter voltage which shows transition from forward active to saturation mode. Differences in current gain were observed between a 2N2222 and 2N3904 transistor. In the common-emitter circuit proper biasing at the q-point was accomplished through choosing resistor values and frequency ranges. Phase shift, saturation, and small signal gain was observed through oscilloscope measurements and provided further examples of alternating current BJT characteristics. This experiment provided a comprehensive understanding of BJT design, operating regions, amplification, and frequency response.

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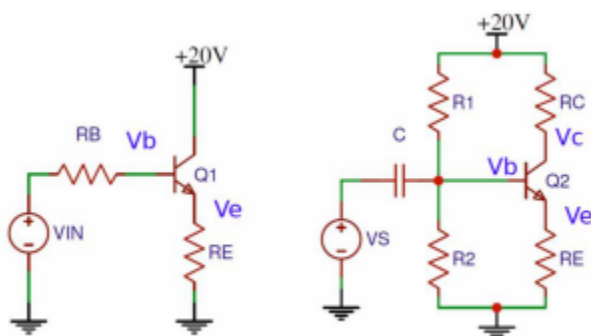
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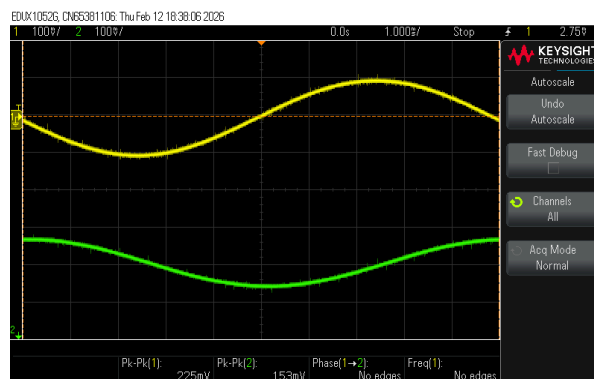
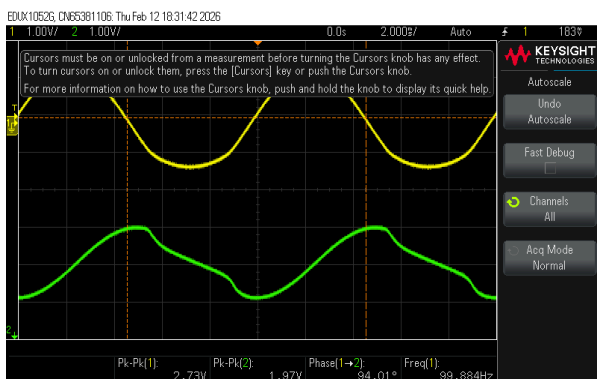
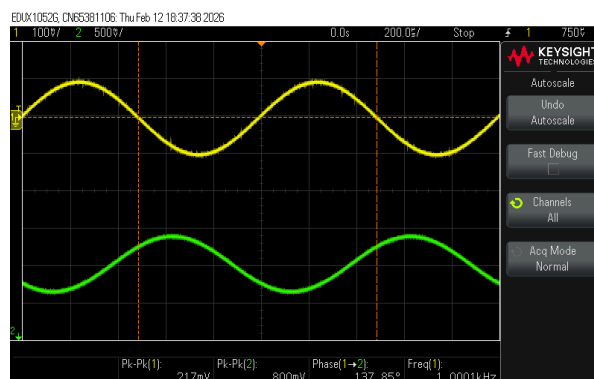
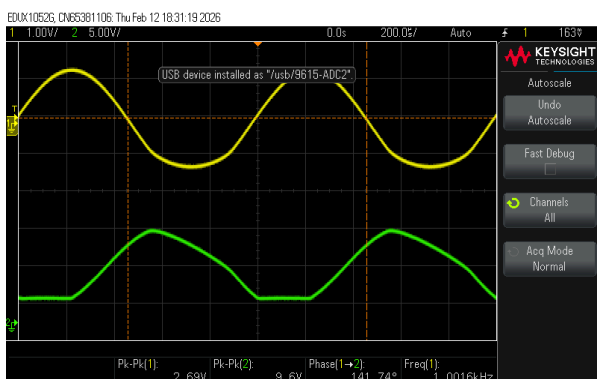
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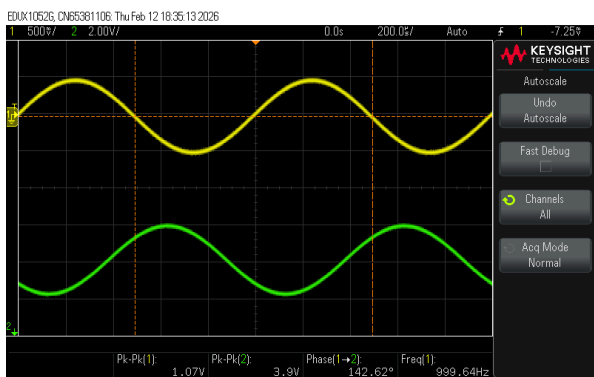
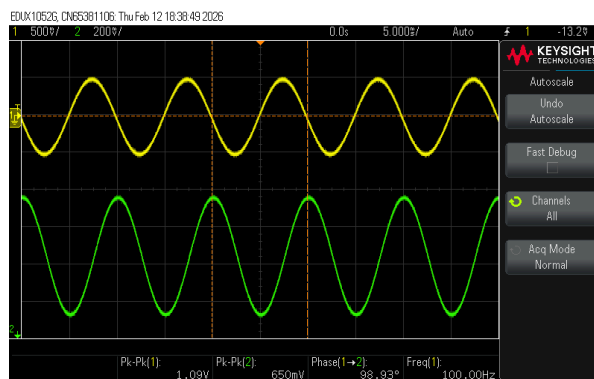
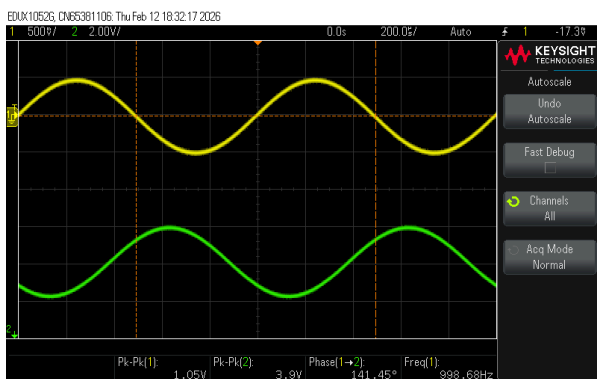
APPENDICES

A. Common-Collector and Common-Emitter BJT Breadboard Layouts



B. All Oscilloscope Readings





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